Not All Experience Belongs in the Weights: Component Routing for Self-Improving GUI Agents

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Paper version: arXiv 2610.01787v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *A trajectory bundles items with different properties, so a conclusion about the bundle depends on its mix.*

Fine-tuning and retrieval each help; the paper asks which **component** of the experience belongs in which place, measured on the same items.

Motivation: three open gaps

1. **Mixed verdicts.** Weights vs prompt studies disagree.
2. **Bundled unit.** Whole trajectories mix item types.
3. **Unpaired tests.** Pipelines differ before destinations do.

Evaluation setup

MobileGym · 1 106 inst. | AndroidWorld · 232 inst. | GUI-Owl-8B | Qwen3-VL-8B | ScaleCUA-7B | 3 seeds | LoRA r 16, 3 epochs | checker, no model judge

- **All online:** held-out success by each environment's checker.
- **Held out:** new templates (practiced apps) and unseen apps.

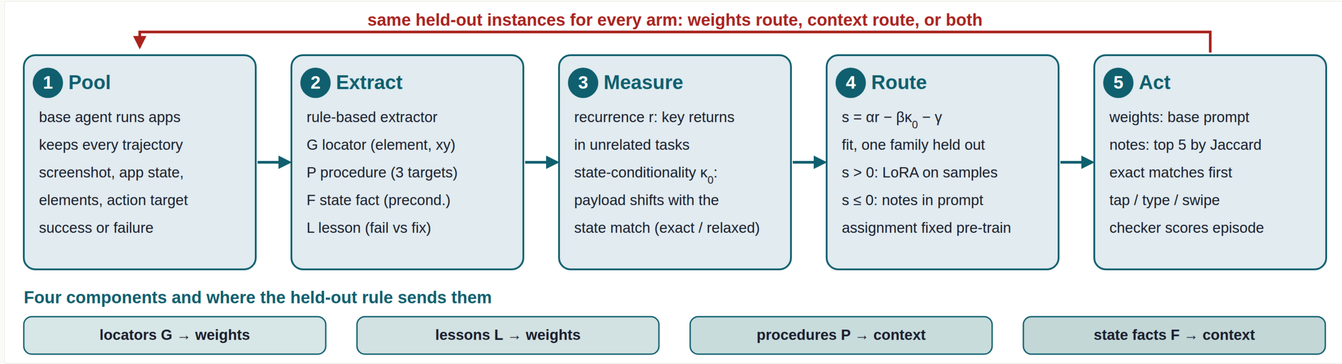
Our read (CUA Radar)

73	★★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out templates; 3 seeds; whole-trajectory baselines; ablation. Budget match, cost: unclear.
- **Flags:** none recorded. Code announced.
- Pattern K27. Authors' numbers, not rerun.

Method: four components, two measured properties, one rule

- **Sees / acts:** screenshot, element list and app state in; each backbone's tap / type / swipe out.
- **Routes:** rule-extracted items become notes in the prompt or LoRA samples.
- **Signal:** checker-scored pool episodes; lessons by preference pairs, rest SFT.



Our redraw (not the paper's figure): pool, extraction, measurement, routing and the held-out run.

$$s_\theta(c) = \alpha\, r(c) - \beta\, \kappa_0(c) - \gamma$$

Rule: s > 0 goes to the weights, else to the context

Key results: MobileGym held-out success (%)

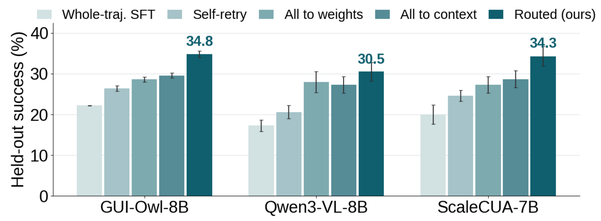


Table 2, redrawn: mean ± std, 3 seeds; online, checker-scored.

Routed exceeds the better single route by +4.4 pts on MobileGym

SO WHAT → *AndroidWorld: see controls table.*

Interventions: gain over base agent (points)

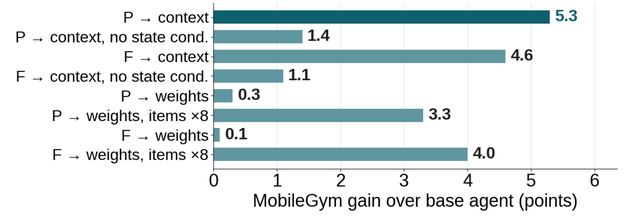


Table 1, GUI-Owl-8B column, redrawn. P procedures, F state facts; one property changed per row.

SO WHAT → *Dropping the state condition removes most of the context gain. Copying items ×8 moves procedures from −4.2 to −0.8 route gain: toward the weights, still below zero on average.*

Headline Numbers

+3.5
pts vs best single dest.
avg., 3 backbones × 2 envs

+4.4 MobileGym avg.
+9.3 vs best whole-traj.
24/24 rule sign, held-out

The paper's own controls

CONTROL	RESULT	COMPARED WITH
AndroidWorld (G / Q / S)	33.3 / 33.2 / 32.6	all-ctx 33.6 / 28.0 / 29.0
Reverse assign., MobileGym	25.2 / 24.4 / 23.2	routed 34.8 / 30.5 / 34.3
Rule fit, held-out family	24 of 24 cells	19 with CI clear of zero
Training, GUI-Owl MobileGym	routed 6.1 GPU-h	all-weights 15.4 GPU-h

Takeaway

Routing by recurrence and state-conditionality beat whole-trajectory baselines; on AndroidWorld one cell ties.

3 backbones, 2 envs | 3 seeds | 1.07× prompt tokens | 24 of 24 cells

Scan to read the full paper

arxiv.org/abs/2610.01787v1